

Homework 5

● Graded

Student

Myles Sartor

Total Points

37 / 37 pts

Question 1

Problem 1

8 / 8 pts

✓ - 0 pts Correct

- 1 pt (a) $X \sim \text{Geom}(p=0.1)$
- 1 pt (a) correct summation
- 1 pt (a) correct derivation of the summation using appropriate rules
- 1 pt (a) answer: $\frac{10}{19}$

(b)

- 1 pt (b) $Y \sim \text{NBin}(r=2, p=0.1)$
- 1 pt (b) correct summation
- 1 pt (b) correct derivation of the summation using appropriate rules
- 1 pt (b) answer: $\frac{181}{361}$

- 8 pts Incorrect

Question 2

Problem 2

6 / 6 pts

✓ - 0 pts Correct

- 0.5 pts correct pdf for X : $P(X = k) = (2/3)^k (1/3)$
- 0.5 pts correct pdf for Y : $P(Y = k) = e^{-3} \frac{3^k}{k!}$
- 1 pt should use the fact that X and Y are independent, then $P(X=x, Y=y) = P(X=x)P(Y=y)$
- 2 pts correct set up: $P(X = Y) = \sum_{k=1}^{\infty} P(X = k)P(Y = k)$
- 1 pt reduce $P(X = Y) = \sum_{k=1}^{\infty} P(X = k)P(Y = k)$ correctly to $\frac{e^{-3}}{2} \sum_{k=1}^{\infty} \frac{2^k}{k!}$
- 0.25 pts recognize that $\sum_{k=1}^{\infty} \frac{2^k}{k!} = e^2 - 1$
- 0.25 pts correct simplified answer should be $\frac{e^{-1}-e^{-3}}{2}$
- 6 pts incorrect/missing solution
- 0.25 pts algebraic error

Question 3

Problem 3

6 / 6 pts

✓ - 0 pts Correct

- 0.5 pts part a: minor error in expression
- 1 pt part a: major error in expression
- 0.5 pts part b: minor error in expression
- 1 pt part c: error in either variance or expected value
- 2 pts part c missing
- 6 pts missing

Question 4

Problem 4

6 / 6 pts

✓ - 0 pts Correct

- 1 pt (a) $P(-3 < X - 1 < 3)$
- 1 pt (a) correct standardization
- 1 pt (a) answer: 0.475

(b)

- 1 pt (b) $P(1 < X < 2, -2 < X < -1)$
- 1 pt (b) correct standardization
- 1 pt (b) answer using CDF of standard normal

- 6 pts Incorrect

Question 5

Problem 5

5 / 5 pts

✓ - 0 pts Correct

- 1 pt $P(X=3)$ incorrect
- 0.5 pts d asks for $P(\leq 3)$
- 0.5 pts e asks for $P(X > 3)$
- 0.75 pts major error in part d
- 0.5 pts minor error in e/f
- 3 pts missing c, d, e
- 1 pt missing part e
- 5 pts no work

✓ - 0 pts Correct

- 1 pt (a) $\lambda = 0.4 * 10 = 4$ for 10-page lecture

- 1 pt (a) $P(Y = 4) = e^{-4} \left(\frac{4^4}{4!} \right) = e^{-4} \left(\frac{32}{3} \right)$

- 1 pt (b) $P(\text{no typo in 1 page}) = P(X = 0) = e^{-0.4}$

- 1 pt (b) $P(\text{number of no typo pages in 10} \geq 1) = 1 - P(\text{number of no typo pages in 10} = 0) = 1 - (1 - e^{-0.4})^{10}$

- 1 pt (c) $P(\text{number of no typo pages} = 3) = \binom{10}{3} (e^{-0.4})^3 (1 - e^{-0.4})^7$

- 6 pts incorrect/missing solution

for $|x| < 1$ } of infinite
 Geometric Series: $S = \frac{a}{1-r}$, $r < 1$.
 • $a = p$ → First term
 • $r = q^2$ → Common ratio

Homework 5

1. $p = 0.10 \rightarrow \text{Success}$ } → Geometric Distribution } $P(X=K) = p(1-p)^{K-1}$, $K=1, 3, 5, \dots$
 $1-p = 0.90 \rightarrow \text{Failure}$ } $(1-p) = q \rightarrow pq^{K-1}$
 ↳ First success in a series of independent Bernoulli trials. ↳ $1-q = p$

$$a) P = P(X=1) + P(X=3) + P(X=5) + P(X=7) \dots \rightarrow P = pq^{1-1} + pq^{3-1} + pq^{5-1} + pq^{7-1} \dots$$

$$P = \frac{p}{1-q^2} \rightarrow \frac{p}{(1+q)(1-q)} = \frac{p}{p(1+q)} = \frac{1}{1+q} = \frac{1}{1+0.9} = \frac{1}{1.9} = \boxed{0.526}$$

Or

$$\sum_{k=1}^{\infty} pq^{k-1} = 0.10(0.90)^{1-1} = 0.10, 0.10(0.90)^{3-1} = 0.081$$

★ Use first few (2) terms to calculate:

$$\frac{a_1}{1-r} \rightarrow \frac{0.10}{1-0.081} = \frac{0.1}{0.919} = 0.526 \text{ or } \boxed{\frac{10}{19}}$$

→ Final answer: $\boxed{\frac{181}{361}}$

b) Negative binomial (# of trials until r -th success). r successes

$$P(X=K) = \binom{K-1}{r-1} p^r (1-p)^{K-r}, K=2, 4, 6, \dots$$

$2K \rightarrow \text{even calls}$ ↳ $q = 1-p$

$$P(X=2K) = \binom{2K-1}{2-1} p^2 (1-p)^{2K-2} \rightarrow \binom{2K-1}{2-1} p^2 q^{2K-2}$$

$$P = \sum_{k=1}^{\infty} p^2 \binom{2K-1}{2-1} q^{2K-2}$$

$$= p^2 \sum_{k=1}^{\infty} (2K q^{2K-2} - q^{2K-2}) = p^2 \left(2 \sum_{k=1}^{\infty} K q^{2K-2} - \sum_{k=1}^{\infty} q^{2K-2} \right)$$

Sum one: $\sum_{k=1}^{\infty} K q^{2K-2} = \sum_{k=1}^{\infty} K (q^2)^{K-1} \rightarrow \frac{1}{(1-q^2)^2}$

★ Sum two: $\sum_{k=1}^{\infty} q^{2K-2} = \sum_{k=0}^{\infty} (q^2)^K = \frac{1}{1-q^2}$

$$1-q^2 = (1-q)(1+q) = p(1+q)$$

Since $p = 1-q$

$$(1-q^2)^2 = p^2(1+q)^2$$

Combined: $P = p^2 \left(2 \cdot \frac{1}{(1-q^2)^2} - \frac{1}{1-q^2} \right) \rightarrow p^2 \cdot \frac{1}{1-q^2} \left(\frac{2}{1-q^2} - 1 \right) \rightarrow p^2 \cdot \frac{1}{1-q^2} \left(\frac{2-(1-q^2)}{1-q^2} \right)$

$$= p^2 \cdot \frac{1}{1-q} \left(\frac{1+q^2}{1-q^2} \right) = \frac{p^2(1+q^2)}{(1-q^2)^2} = \frac{p^2(1+q^2)}{p^2(1+q)^2} = \frac{1+q^2}{(1+q)^2} = \frac{1+0.9^2}{(1+0.9)^2} = \frac{1.81}{3.61} = \frac{181}{361} = 0.501$$

• Known

Taylor series: $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

Hw 5

$$2. p = \frac{1}{3}, \lambda = 3$$

$\hookrightarrow$ Geometric $\hookrightarrow$ Poisson

$$P(X=K) = (1-p)^{K-1} p, K=1, 2, 3, \dots \rightarrow P(X=K) = \left(\frac{2}{3}\right)^{K-1} \cdot \frac{1}{3}$$

$$P(Y=K) = \frac{e^{-\lambda} \lambda^K}{K!}, K=0, 1, 2, \dots \rightarrow P(Y=K) = \frac{e^{-3} 3^K}{K!}$$

$$P(X=Y) = \sum_{K=0}^{\infty} P(X=K, Y=K)$$

$$\hookrightarrow P(X=Y) = \sum_{K=1}^{\infty} P(X=K) \cdot P(Y=K)$$

$$\hookrightarrow P(X=Y) = \sum_{K=1}^{\infty} \left(\frac{2}{3}\right)^{K-1} \cdot \frac{1}{3} \cdot \frac{e^{-3} 3^K}{K!}$$

$$= \frac{1}{3} e^{-3} \sum_{K=1}^{\infty} \left(\frac{2}{3}\right)^{K-1} \frac{3^K}{K!}$$

$$= \frac{1}{3} e^{-3} \sum_{K=1}^{\infty} \left(\frac{2}{3}\right)^{K-1} \frac{2 \cdot 3^{K-1}}{K!}$$

$$= e^{-3} \sum_{K=1}^{\infty} \frac{2^{K-1} \cdot 3^{K-1}}{K!}$$

$$= e^{-3} \sum_{K=1}^{\infty} \frac{6^{K-1}}{K!}$$

$$P(X=Y) = e^{-3} \sum_{m=0}^{\infty} \frac{6^m}{(m+1)!}$$

$$\frac{6^m}{(m+1)!} = \frac{1}{6} \cdot \frac{6^{m+1}}{(m+1)!}$$

Right here

the coefficient is $\frac{6}{1}$, we need $\frac{1}{1}$ for

the known series estimation.

$$P(X=Y) = \frac{e^{-1} - e^{-3}}{2} \approx 0.159$$

Index changes to zero b/c we're adding in the one now

$\frac{6}{6} = 1$, which is the coefficient in $\frac{x^n}{n!}$ form!

$$P(X=Y) = \frac{e^{-3}}{6} \sum_{m=0}^{\infty} \frac{6^m}{(m+1)!}$$

→ Add in the

entire expression and

factor $\frac{1}{6}$ constant out

before manipulation.

$$P(X=Y) = \frac{e^{-3}}{6} \sum_{n=1}^{\infty} \frac{6^n}{n!}$$

subbed in

n to make it a

known Taylor

series, index changes back

to one to account

for m+1 being taken

out for n.

Subtract one

b/c the sum

started at

n=1, while the

Taylor series for

e^x starts at

n=0. We want to

exclude the n=0 term in the

Taylor series for e^x .

$$3^K = 3 \cdot 3^{K-1}$$

• sub out K for

m+1 to get rid of -1 in exponent and change index to start at zero. Now

its easier to work with and simplify to a known series!

- girls in any order, and then a girl must be the 5th child for it to end with 3 girls and 2 boys!!
- 50/50 to have a boy/girl
 - $P(\text{Boy}) = p = 0.5$
 - $P(\text{girl}) = 1 - p = 0.5$

Hw 5. Con.

3. a) $P(K \text{ successes in } n \text{ trials}) = \binom{n}{K} p^K (1-p)^{n-K}$ ↗ binomial distribution

$P(2 \text{ boys and } 3 \text{ girls}) = \binom{5}{2} (0.5)^2 (0.5)^{5-2}$, $p = 0.5$

$$= \binom{5}{2} (0.5)^2 (0.5)^3$$

$$= \boxed{\binom{5}{2} (0.5)^5}$$

$$\binom{5}{2} = \frac{5!}{2!3!} = 10$$

$$10 \cdot \left(\frac{1}{2}\right)^5 = 10 \cdot \frac{1}{32} = \frac{10}{32} = \boxed{\frac{5}{16}}$$

↗ can also picture as binomial! → $\binom{4}{2} (0.5)^2 (0.5)^{4-2} \cdot 0.5$

b) Negative Binomial (Have children until they have 3 girls, whilst having 2 boys along the way to this outcome).

$P(2 \text{ Boys on the way to } 3 \text{ girls}) = \binom{5-1}{3-1} (0.5)^2 (0.5)^{5-3} \cdot 0.5$ ↗ 5th girl

↳ $P(2 \text{ Boys and } 2 \text{ girls in any order but the 5th child must be a girl})$

$$= \binom{4}{2} (0.5)^2 (0.5)^2 \cdot 0.5$$

$$= 6 \cdot (0.5)^5 = 6 \cdot \left(\frac{1}{2}\right)^5$$

$$= 6 \cdot \frac{1}{32} = \boxed{\frac{3}{16}}$$

c) $E(C) = \frac{r}{p} = \frac{3}{0.5} = \frac{3}{\frac{1}{2}} = \boxed{6}$

$$\text{Var}(C) = \frac{r(1-p)}{p^2} = \frac{3(0.5)}{(0.5)^2} = \frac{3}{0.5} = \frac{3}{\frac{1}{2}} = \boxed{6}$$

$$|x-1| < 3 \Rightarrow -3 < x-1 < 3$$

$$z = \frac{x - \mu}{\sigma}$$

Hw 5.

$$4. E[X] = 4, \text{Var}[X] = 9$$

$$N(\mu, \sigma^2)$$

$$N(4, 9)$$

$$Z = \frac{X - 4}{3}$$

$$a) P(|x-1| < 3) = P(-3 < x-1 < 3) = P(-2 < x < 4)$$

$$P(-2 < x < 4) = P\left(\frac{-2-4}{3} < Z < \frac{4-4}{3}\right)$$

$$= P(-2 < Z < 0) = P(0 < Z < 2) \rightarrow \text{Symmetry}$$

$$= \Phi(2) - \Phi(0) \text{ or } \Phi(0) - \Phi(-2)$$

$$= 0.9772 - 0.5 = 0.4772$$

$$b) P(1 < x^2 < 4) \rightarrow P(1 < x < 2) \text{ or } P(-2 < x < -1)$$

$$\hookrightarrow P\left(\frac{1-4}{3} < Z < \frac{2-4}{3}\right) \hookrightarrow P\left(\frac{-2-4}{3} < Z < \frac{-1-4}{3}\right)$$

$$\hookrightarrow P(-2 < Z < -\frac{5}{3}) \hookrightarrow P(-2 < Z < -\frac{5}{3})$$

$$P(-2 < Z < -\frac{5}{3}) + P(-1 < Z < -\frac{2}{3}) \text{ or } P(\frac{5}{3} < Z < 2) + P(\frac{2}{3} < Z < 1)$$

$$\Phi(-\frac{5}{3}) - \Phi(-2) + \Phi(-\frac{2}{3}) - \Phi(-1) \text{ or } \Phi(2) - \Phi(\frac{5}{3}) + \Phi(1) - \Phi(\frac{2}{3})$$

$$0.0247 + 0.0927 = 0.1174$$

$$\hookrightarrow 0.9772 - 0.9525 + 0.8413 - 0.7486$$

$$= 0.0247 + 0.0927 = 0.1174$$

5. Poisson Distribution: $P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}$, $k=0,1,2,\dots$

$\lambda = 2.5 \rightarrow$ All happening in one minute so rate stays the same (No multiplying by min.)

$$a) P(x=0) = \frac{e^{-2.5} 2.5^0}{0!} = \boxed{e^{-2.5}} = 0.082$$

$$b) P(x=1) = \frac{e^{-2.5} 2.5^1}{1!} = \boxed{2.5e^{-2.5}} = 0.205$$

$$c) P(x=2) = \frac{e^{-2.5} 2.5^2}{2!} = \boxed{\frac{6.25e^{-2.5}}{2}} = 0.256$$

$$d) P(x \leq 3) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$$

$$P(x=3) = \frac{e^{-2.5} 2.5^3}{3!} = \frac{15.625e^{-2.5}}{6} = 0.214$$

$$P(x \leq 3) = \sum_{k=0}^3 \frac{e^{-\lambda} \lambda^k}{k!} = 0.082 + 0.205 + 0.256 + 0.214 = \boxed{0.757}$$

$$f) P(x > 3) = 1 - P(x \leq 3) = 1 - 0.757 = \boxed{0.243} \rightarrow$$

$$1 - \sum_{k=0}^3 \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{or} \quad \sum_{k=3}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!}$$

for more than 1 success
NB: # of trials until r successes $\rightarrow$ until keyword
B: # of successes in n trials for NB and Geometric
Use for $\rightarrow$ Geometric:
1 success $\rightarrow$ 100

Hw 5 Con.

6. Poisson: $P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}$

$\lambda = 0.4$

$\lambda \cdot 10 \text{ pages} = 0.4 \cdot 10 = 4$

a) $P(X=4) = \frac{e^{-4} 4^4}{4!} = \frac{e^{-4} \cdot 256}{24} = 0.192$

$\hookrightarrow$ 10 page rec.
not one page

$\rightarrow$ Complements
flip both typo and page count statements

* b) $P(0 \text{ typos on 1 page}) = e^{-\lambda} = e^{-0.4} = 0.670$

$P(\text{at least 1 typo per page}) = 1 - P(0 \text{ typos per page}) \rightarrow \text{complement}$
 $\hookrightarrow 1 - e^{-\lambda} \rightarrow 1 - e^{-0.4} = 0.33$

$P(\text{all 10 pages have at least one typo}) = (0.33)^{10} = (1 - e^{-0.4})^{10}$

$P(\text{at least 1 page with no typos}) = 1 - (1 - e^{-0.4})^{10}$

$\hookrightarrow 1 - P(\text{all 10 pages have at least one typo}) = 1 - 0.00015 = 0.99985$

or
 $\hookrightarrow 1 - P(\text{no pages without a typo}) \rightarrow 1 - (1 - e^{-0.4})^{10} \rightarrow \text{account for 10 pages}$

* $p = e^{-\lambda}$
success is a page with no typos!

(c) Binomial, each page is a trial:

$P(Y=3) = \binom{10}{3} (e^{-0.4})^3 (1 - e^{-0.4})^7 = 0.0154$

$\rightarrow$ 3 successes in 10 trials, binomial

* $K=3$

